THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Fish breathe through: 1

(a) gills **(b)** lungs **(c)** spiracles **(d)** skin only

2. The cracking of the voice in boys is due to growth of the: 1

(a) voice box (larynx) **(b)** heart

(c) lungs **(d)** stomach

3. In convection, heat is carried by: 1

(a) moving particles of the fluid **(b)** waves in space

(c) light only **(d)** vibrating solid

4. The SI unit of speed is: 1

(a) kilometre **(b)** kilogram **(c)** second **(d)** metre per second

5. The process of breaking down complex food into simple substances is: 1

(a) respiration **(b)** digestion **(c)** egestion **(d)** absorption

6. In general, the growth spurt starts earlier in: 1

(a) girls **(b)** boys

(c) both at exactly the same age **(d)** neither of them

7. In summer we prefer to wear: 1

(a) light-coloured clothes **(b)** dark-coloured clothes

(c) leather clothes **(d)** woollen clothes

8. The instrument in a vehicle which shows the speed is the: 1

(a) odometer **(b)** compass **(c)** clock **(d)** speedometer

9. The largest gland in the human body is the: 1

(a) liver **(b)** thyroid **(c)** salivary gland **(d)** pancreas

10. Adolescence is also called the period of: 1

(a) old age **(b)** teenage **(c)** infancy **(d)** toddlerhood

11. Transfer of heat through a solid without movement of the solid is: 1

(a) convection **(b)** evaporation **(c)** radiation **(d)** conduction

12. The time taken to complete one oscillation of a pendulum is its: 1

(a) speed **(b)** frequency **(c)** amplitude **(d)** time period

13. Humans breathe in air through the: 1

(a) nostrils **(b)** eyes **(c)** ears **(d)** mouth only

14. The enlarged voice box in boys is visible as: 1

(a) tongue **(b)** thyroid **(c)** trachea **(d)** Adam's apple

15. Which of these needs no material medium? 1

(a) all three **(b)** radiation **(c)** convection **(d)** conduction

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do we wear light-coloured clothes in summer? 2

17. A boy runs 400 m in 80 s. Find his speed. 2

18. What are villi? Why are they present? 2

19. Why should adolescents keep themselves clean? 2

20. Why are cooking utensils made of metal? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain how the heart pumps blood in the body. 3

22. Why is physical exercise important for adolescents? 3

23. Why do birds fluff their feathers in cold weather? 3

24. A cyclist rides at 18 km/h. How far does he go in 20 minutes? 3

OR

Explain how cockroach and fish breathe. 3

25. Describe the role of the pituitary gland in puberty. 3

OR

Explain how convection currents are set up in water heated in a beaker. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the role of hormones in growth and mention the diseases linked with hormone problems. 5

OR

Differentiate between conduction, convection and radiation on five points. 5

27. Describe the oscillatory motion of a pendulum and explain terms: oscillation, time period, amplitude. 5

OR

Describe the human excretory system and the formation of urine. 5

END OF QUESTION PAPER — Total: 50 Marks